

Implementation of a Secure IIoT Gateway: 10BASE-T1L Interface for ESP32 Using ADIN1110

*Project conducted within the scope of "Schaltungstechnik 2"

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Abstract—The rise of Industry 4.0 requires robust, long-range communication protocols that extend beyond the capabilities of standard Wi-Fi or Bluetooth. Single Pair Ethernet (SPE), specifically the 10BASE-T1L standard, offers a solution with cable reaches up to 1000 meters. This paper details the design and implementation of a hardware interface connecting an ESP32 microcontroller to the Analog Devices ADIN1110 MAC-PHY. We present a custom PCB design utilizing a galvanic isolation transformer to ensure safety against Power over Ethernet (PoE) faults. Furthermore, a platform-independent C++ driver architecture with a custom Layer 2 protocol was developed. The system was successfully validated, achieving stable 10 Mbps link speeds and demonstrating a real-time, web-controlled PWM actuator application via a secure gateway architecture.

Index Terms—ESP32, ADIN1110, 10BASE-T1L, Single Pair Ethernet, SPI, Industrial IIoT, Gateway

I. INTRODUCTION

In the landscape of the Industrial Internet of Things (IIoT), the "last mile" connectivity for sensors poses a significant challenge. While wireless solutions like the ESP32 are versatile, they suffer from range limitations and susceptibility to interference in industrial environments. Conversely, standard Ethernet requires complex cabling and high component costs.

The IEEE 802.3cg (10BASE-T1L) standard addresses this by enabling 10 Mbps Ethernet over a single twisted pair with a reach of up to 1 km. This project aims to bridge the gap between low-cost microcontrollers and industrial SPE by integrating the Espressif ESP32 with the ADIN1110 MAC-PHY.

A key contribution of this work is the implementation of a "Security Gateway" architecture. By hosting a local Web Server on the ESP32 (Transmitter) and relaying commands over SPE to a passive Receiver, the physical machine actuators are air-gapped from the wireless network, enhancing security in industrial control scenarios.

II. SYSTEM ARCHITECTURE

A. Hardware Design

The system creates a bridge between the wireless capabilities of the ESP32 and the wired reliability of SPE.

1) *Microcontroller Unit (MCU)*: The host controller is an ESP32-DevKitC. It is responsible for the application logic, hosting the WiFi Access Point, and managing the SPI bus. The ESP32 3.3V logic rail supplies the peripheral PHY I/O, ensuring signal level compatibility without level shifters.

2) *PHY Integration (ADIN1110)*: The ADIN1110 was selected for its ultra-low power consumption and integrated MAC, which offloads Ethernet frame processing from the ESP32.

- **Clocking**: The ADIN1110 utilizes an external 25 MHz crystal oscillator for precise timing.
- **Safety Isolation**: A critical design decision was the inclusion of an isolation transformer on the MDI side. This protects the low-voltage ESP32 circuitry from potential high-voltage faults if connected to Power over Ethernet (PoE) segments.

B. Hardware Configuration Settings

To ensure the correct operation of the ADIN1110 MAC-PHY within the designed circuit, specific hardware jumpers must be configured. Table I details the function of each jumper and the operational state validated in this implementation.

TABLE I
PCB JUMPER CONFIGURATION

Jumper	Function Description	Setting
JP1	Power Supply Enables power to the module.	High (True)
JP2	SPI_CFG1 Determines SPI protocol mode.	Low (False)
JP3	TX2P4_EN High: 1.0 V p-p only. Low: Supports 1.0 V & 2.4 V p-p.	Low (False)
JP4	SWPD_EN Software Power-Down Config. Low: Enter power-down on reset.	High (True)
JP5	MS_SEL Master/Slave Selection. High: Prefer Master. Low: Slave.	High (True)

As shown in Table I, the system is configured to support dual voltage amplitudes (JP3 Low) and prefers the Master role (JP5 High), which is typical for a gateway controller.

III. SOFTWARE IMPLEMENTATION

A. Driver Architecture: The "Capsule" Model

To maintain code modularity, a "Capsule" driver architecture was implemented. This separates the low-level SPI register access from the high-level application logic. The driver supports the *Generic SPI* protocol of the ADIN1110, managing the CRC appending and frame encapsulation transparently.

```

1  uint32_t readReg(uint16_t regAddr) {
2      uint16_t header = 0x8000 | (regAddr & 0x1FFF);
3      // Header setup with CRC calculation
4      uint8_t headBuf[2] = {
5          (uint8_t)(header >> 8),
6          (uint8_t)(header & 0xFF)
7      };
8
9      adinSpi.beginTransaction(SPISettings(1000000,
10         ↪ MSBFIRST, SPI_MODE0));
11     digitalWrite(PIN_CS, LOW);
12     // Full duplex transfer handling...
13     digitalWrite(PIN_CS, HIGH);
14     adinSpi.endTransaction();
15     return result;
16 }

```

Listing 1. Low-Level Register Access Implementation

As shown in Listing 1, the driver manages the Chip Select (CS) assertion and SPI transaction integrity, including the specific 2-byte header required by the ADIN1110.

B. Custom Layer 2 Protocol

To minimize latency and overhead, a custom raw Ethernet protocol (EtherType 0x88B5) was devised rather than a full TCP/IP stack on the receiver side. The payload utilizes a lightweight sequence command structure:

- **Format:** SEQ:Duration,Brightness, Duration, Brightness...
- **Logic:** This protocol defines an animation sequence where the receiver interpolates the LED brightness over the specified duration.

C. Fading Engine and PWM Control

The receiver implements a non-blocking state machine that parses the incoming sequence. It utilizes the ESP32's `ledc` hardware peripheral to generate high-resolution PWM signals. A linear interpolation algorithm allows for smooth transitions (fading) between brightness states, enabling complex visual patterns like "breathing" or "strobe" effects.

D. Web Interface (AJAX)

The sender unit hosts a WiFi Access Point and an asynchronous web server. The frontend utilizes the JavaScript Fetch API (AJAX) to send commands to the ESP32 without reloading the webpage. This provides a responsive user experience for manual brightness control and preset selection.

IV. RESULTS AND VALIDATION

A. Hardware Debugging and Via Repair

During the initial assembly, the system failed to establish a link. A deep analysis of the PCB revealed a manufacturing defect: a non-conductive buried VIA on the carrier board, severing the SPI MISO line. After manually bridging this connection, full duplex communication was established. This highlights the importance of electrical continuity testing in prototype PCBs.

B. Signal Integrity and Speed Verification

To verify the signal quality and the achieved data rate on the physical layer, an eye diagram measurement was conducted using an Agilent DSO-X 2002A digital storage oscilloscope. The probe was connected to the TX pins of the ADIN1110.

Fig. 1 shows the captured eye diagram. The analysis of the waveform reveals stable plateaus with a duration (T_{sym}) of approximately 133 ns per symbol.

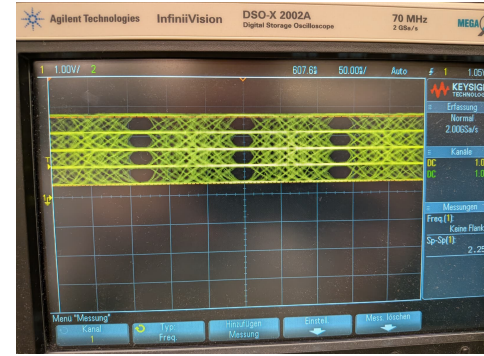


Fig. 1. Eye diagram measured at the TX interface. The five distinct voltage levels which most likely result from reflection showcase the transmission via PAM3 modulation.

The resulting symbol rate (Baud rate) f_{sym} is calculated as:

$$f_{sym} = \frac{1}{T_{sym}} \approx \frac{1}{133 \times 10^{-9} \text{ s}} \approx 7.5 \text{ MHz} \quad (1)$$

This result aligns perfectly with the **ADIN1110 datasheet** and the **IEEE 802.3cg** standard, which specify a symbol rate of 7.5 MBd for 10BASE-T1L. To achieve the full 10 Mbit/s throughput, the standard utilizes **PAM3 (Pulse Amplitude Modulation 3)** encoding.

As clearly visible in Fig. 1, the signal utilizes three distinct voltage levels (+1, 0, -1). This observation definitively confirms that the ADIN1110 is correctly operating in PAM3 mode. While the eye diagram also exhibits intermediate artifacts, likely caused by reflections or backwards signaling in the test setup, the presence of the three primary and two secondary (intermediate) levels and the correct timing interval validates the successful physical layer integration.

C. Transmission Performance

Link stability was verified over a 20-meter twisted pair cable. The ADIN1110 successfully negotiated a 10 Mbps link. Throughput tests using the raw frame protocol confirmed that

the SPI bus (running at 1 MHz) was the bottleneck, not the SPE link, which is sufficient for control applications. The speed of transition therefore can be assumed to depend heavily on the ESP32s speed of writing into the ADIN1110s registers.

D. Application Demonstration

To demonstrate the system's capability, a "Remote Industrial Actuator" scenario was constructed.

- 1) **User Input:** A user connects to the ESP32 WiFi and selects a "Heartbeat" pattern via the web interface (Fig. 2).
- 2) **Processing:** The Sender ESP32 translates this into the raw SEQ protocol.
- 3) **Transmission:** The frame is transmitted via 10BASE-T1L to the Receiver unit.
- 4) **Actuation:** The Receiver parses the frame and executes the smooth LED fading sequence (Fig. 3).

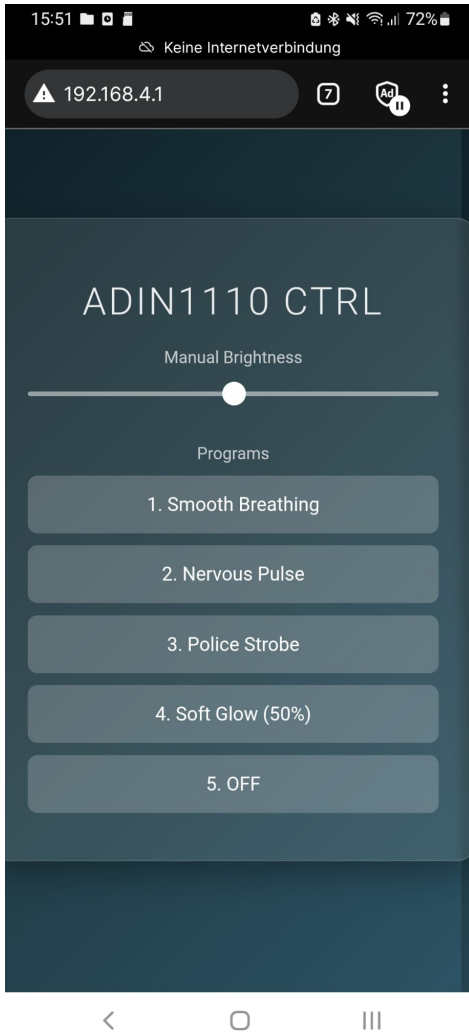


Fig. 2. The responsive Web Interface hosted on the Sender ESP32, allowing manual PWM control and preset selection.

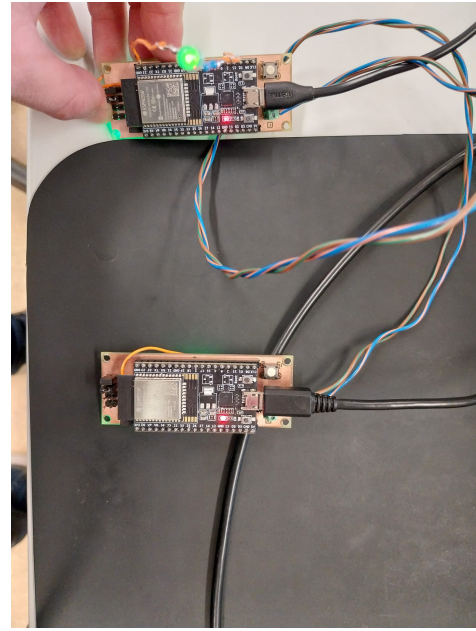


Fig. 3. The Receiver unit executing a PWM fading sequence transmitted via the 10BASE-T1L link.

V. CONCLUSION

This paper presented the successful design of a 10BASE-T1L node based on the ESP32. The hardware integration, specifically the safety-oriented transformer coupling, proved robust after resolving manufacturing defects. The developed firmware architecture demonstrates that low-cost microcontrollers can effectively drive industrial SPE interfaces. The implementation of a custom Layer 2 protocol and a web-based control gateway illustrates the potential of this technology for secure, air-gapped industrial control systems.

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